

resource report and netlist, which preserves module boundaries, to facilitate datapath analysis for LEC verification between the RTL design and synthesized netlist.

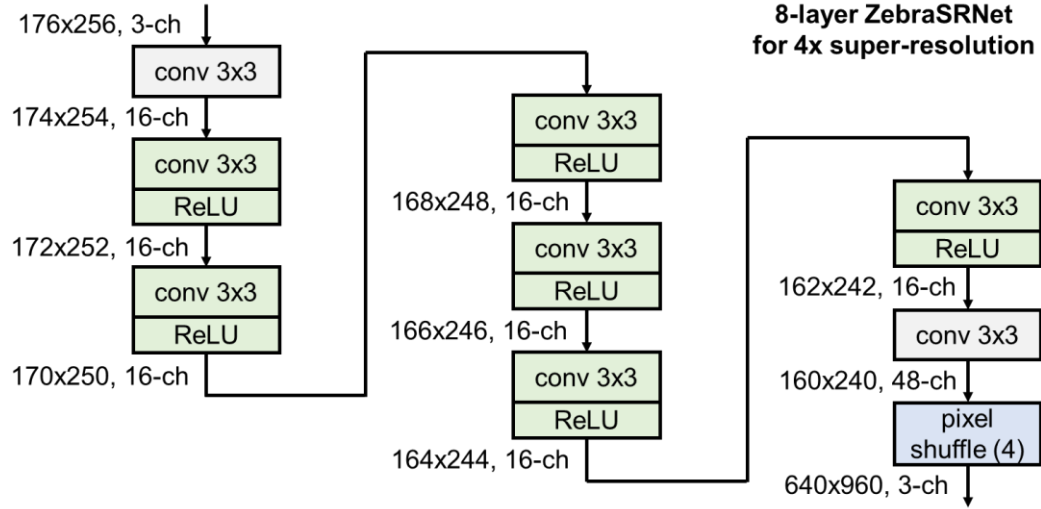


Figure 2. Network architecture of ZebraSRNet.

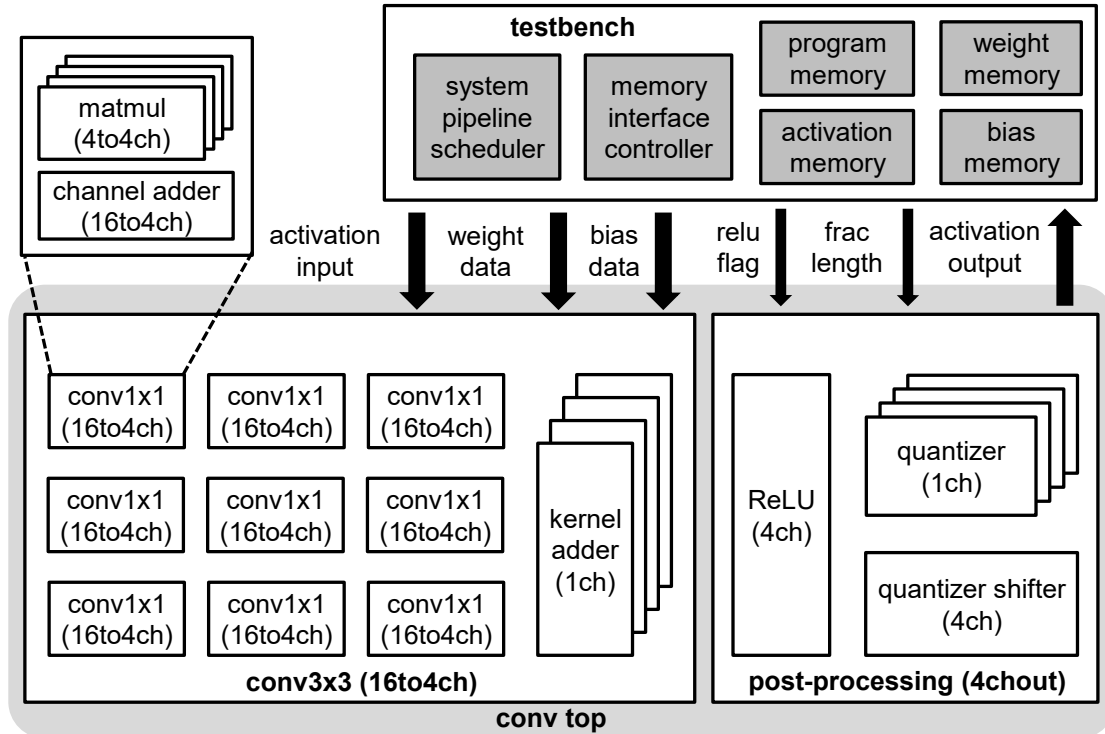


Figure 3. Design overview of ZebraSRNet CNN accelerator.

In this lab, we utilize a highly-parallel CNN accelerator to demonstrate the practical effectiveness of MDP flow. Figure 2. shows an 8-layer CNN – ZebraSRNet for four times super-resolution and its basic operations. Then, a customized ZebraSRNet CNN accelerator is implemented with 8-bit dynamic fixed-point quantization scheme. The hardware architecture and I/O definition are shown in Figure 3. and Table 1., respectively. The accelerator can support convolution computations with 16-channel (3×3 input) to 4-channel (1×1 output) per cycle. Hence, there are a total of $16 \cdot 4 \cdot 3 \cdot 3 = 576$ multiply-accumulate operations (MACs) in the design.

Signal	Description
clk	Clock.
rst_n	Active low reset.
din_enable	Enable signal of input data.
dout_valid	Valid signal of output data.
in_activation	Input feature map of a $16 \times 3 \times 3$ tensor ($ch_{in} \times h \times w$).
weight	Weight of a $4 \times 16 \times 3 \times 3$ tensor ($ch_{out} \times ch_{in} \times h \times w$).
bias	Bias of 4 output channels
out_activation	Output feature map of a $4 \times 1 \times 1$ tensor ($ch_{out} \times h \times w$).
in_fl	Fractional length of input feature map.
wgt_fl	Fractional length of weight.
bias_fl	Fractional length of bias.
out_fl	Fractional length of output feature map.
relu	With or without (1/0) ReLU function.

Table 1. I/O definition of ZebraSRNet CNN accelerator.

Action Items

Please record the runtime and fill in the table according to the following experiments.

	Flat Compare (min.)	Hierarchical Compare (min.)
Basic Flow	Abort	
MDP Flow		

I. Basic Flow

Change the directory to `./lec/basic/`, and perform the basic LEC flow from Lab09 in this design.

1. (Optional) Run flat compare.

```
$ lec -xl -nogui -dofile lec_all.do
```

2. Run hierarchical compare.

```
$ lec -xl -nogui -dofile lec_hier_all.do
```

Note: If “abort” is displayed in the terminal and it might result in a comparison failure, you can directly terminate the process. To terminate the process, press “**Ctrl+Z**” to suspend first and type:

```
$ ps
```

You will find the process ID (PID) of LEC. Type the following command to kill the suspended process. (If you perform the LEC flow with `run_lec.sh`, please kill the PID of the shell script.)

PID	TTY	TIME	CMD
4343	pts/1	00:00:01	lec
4440	pts/1	00:00:00	ps
20390	pts/1	00:00:01	tcsh

```
$ kill -9 <PID>
```

II. MDP Flow

Change the directory to `./lec/mdp_flow/`, open **lec_setup.do**, and fill in the path of resource file (in `../syn/report/report_resources_xxx`).

What is in the resource report? _____.

1. Run flat compare.

```
$ lec -xl -nogui -dofile lec_all.do
```

Read the script and understand what you are doing.

2. Run hierarchical compare.

```
$ lec -xl -nogui -dofile lec_hier_all.do
```

Read the script and understand what you are doing.

What is the difference between the basic flow and MDP flow? _____.

What is “quality evaluated xx% success” displayed in LEC output terminal? _____.

Which LEC flow would you choose for a large-scale design? (basic or MDP flow, flat or hierarchical) _____. Why? _____.

Appendix

I. Official MDP Flow

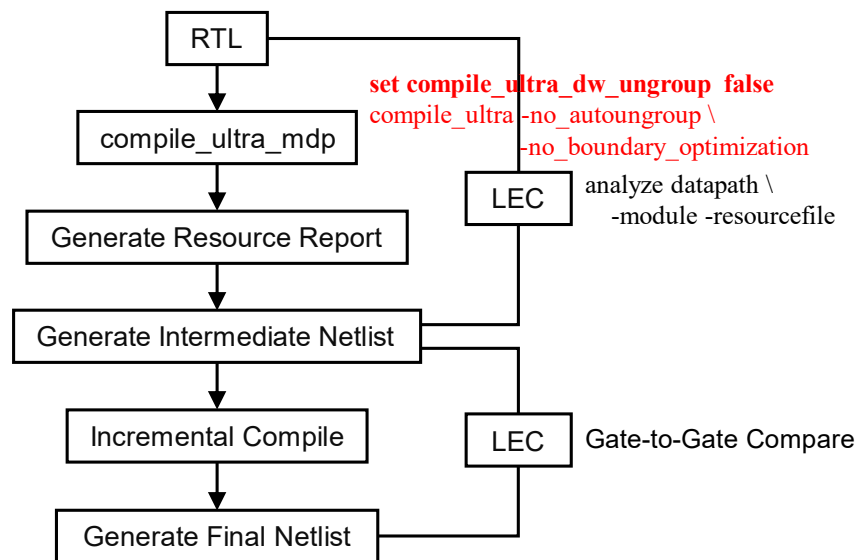


Figure 4. Official MDP flow for LEC verification.

If the current LEC flow still encounters comparison failures, you can consider trying the officially released version of MDP flow by Cadence, as illustrated in Figure 4. The LEC verification is divided into two-step comparisons: RTL to intermediate netlist comparison and intermediate to final netlist comparison. Besides, during the synthesis, DesignWare module boundaries are further to be preserved to help datapath analysis for LEC verification.

II. LEC Debugging

1. After running non-GUI commands, save the session.

```
LEC> save session <name>
```

2. Open GUI, restore the session and switch to LEC mode.

```
$ lec -xl
```

SETUP> restore session <name>

SETUP> set system mode lec (or you can click “LEC” button in GUI)

3. Open “Mapping Manager” and find the problematic key point.
4. Diagnose it via “Diagnosis Manager”.

III. File Structure

Directory	Filename	Description
hdl/	conv_define.v	Parameter define
	conv_top.v	Top module
	conv3x3_16to4ch.v	Convolution 3x3 engine
	conv1x1_16to4ch.v	Convolution 1x1
	matmul.v	Matrix multiplication
	kernel_adder.v	Sum 9 pixel and bias
	postproc.v	Postprocessing
	shifter_decoder.v	Shift amount controller
	ReLU.v	ReLU function
	quantization_1ch.v	Output quantization
sim/	data/	Test pattern
	result/out2img.py	Input image pattern
	run.sh	Shell script
	test_run.f	File list for testbench
	test_cnn.v	Testbench
	task.v	Task definition
syn/	*.tcl & .synopsys_dc_setup	Synthesis scripts
lec/basic	run_lec.sh	Shell script
	golden.f, revised.f	File list for LEC
	*.do	Do files for basic LEC flow
lec/mdp_flow	run_lec.sh	Shell script
	golden.f, revised.f	File list for LEC
	*.do	Do files for MDP flow

Reference

- [1] Conformal Equivalence Checking User Guide